SAP Note

2942344 - SAP Integration Suite - Message Metric Definition

Component: LOD-HCI (OnDemand > SAP Cloud Integration), Version: 19, Released On: 28.08.2024

Symptom

How is a message defined and metered in SAP Integration Suite

Other Terms

Integration Suite, Cloud Integration, API Management,

Reason and Prerequisites

I would like to understand the new message definition and how a message is metered in SAP Integration Suite.

Solution

What is a message, as defined in the service description guide?

A message is an electronic communication exchanged via the capabilities of the Cloud Service. If a message is larger than 250 KB, any amount in excess of 250 KB is charged as one additional message for each 250 KB, or portion thereof.

This is the standard metering metric that is used for metering the usage of all SAP Integration Suite capabilities.

Simplified Metering Explanation

An “integration message” is a message processed through Cloud Integration, API Management, or Open Connectors capabilities of SAP Integration Suite. If an integration message is larger than 250 kilobytes, any amount in excess of 250 kilobytes will be charged as one additional message for each 250 kilobytes or portion thereof. 1 integration message is counted as 1 transaction.

An “edge message” is a message processed through the Edge Integration Cell runtime of SAP Integration Suite. If an edge message is larger than 250 kilobytes, any amount in excess of 250 kilobytes will be charged as one additional message for each 250 kilobytes or portion thereof. 2 edge messages are counted as 1 transaction.

An “event message” is a message processed through the Event Mesh capability of SAP Integration Suite. If an event message is larger than 250 kilobytes, any amount in excess of 250 kilobytes will be charged as one additional message for each 250 kilobytes or portion thereof. 100 event messages are counted as 1 transaction.

- In the context of the ***Cloud Integration*** capability, the combined (aggregated) size of all outbound messages via one or more receiver(s) adapter is considered for message metering. You can find more details with examples in the detailed explanation section. When editing/customizing integration flows (if editable) that are delivered as part of pre-defined integration packages by SAP on SAP Business Accelerator Hub, all messages are counted, even if the corresponding scenarios exclusively represent SAP-to-SAP integration.
- In the context of ***API Management*** and ***Open Connectors*** capabilities, every API call (and payload size) is considered for message metering.
- In the context of ***Event Mesh*** capability, only outbound events are considered for message metering.
- If a message is processed by more than capability of SAP Integration Suite, then the message is charged for each time it is processed by a capability.
- SAP-to-SAP free messages are only offered for the Cloud Integration capability. This is applicable only for the messages processed by prepackaged integration packages published by SAP on the SAP Business Accelerator Hub.

IMPORTANT: Please note that there is no concept of dev, test, and production landscapes in SAP Integration Suite. Messages on all tenants are metered based on the description in this note.

For a detailed explanation of message metering, see below:

Cloud Integration

In the context of Cloud Integration, outbound message size via one or more receiver adapters are considered for message metering.

During every iteration of integration flow execution, sizes of all outbound messages via all receiver adapters are aggregated and considered for message count (subject to exceptions mentioned below). Each aggregated message size of up to 250 KB is calculated as 1 message. If message size is greater than 250 KB, the excess is counted in blocks of 250 KB to determine the total number of messages. If a response is received, message size of both request and response are considered. As a rule of thumb, you can determine the number of messages by dividing the aggregated message size by 250 and rounding up the answer to the next highest whole number.

To help you understand that better, let us consider the following scenarios:

- An integration flow has two receiver adapters. During runtime execution, messages are processed via one receiver adapter and the size of the messages is 50 KB. It is counted as one message because it is less than 250 KB.
- An integration flow has two receiver adapters. During runtime execution, messages are processed via both the receiver adapters and the size of the messages is 50 KB and 40 KB respectively. It is counted as one message because 40 KB +50 KB = 90 KB, which is less than 250 KB.
- An integration flow has two receiver adapters. During runtime execution, messages are processed via both the receiver adapters and the size of the messages is 150 KB and 120 KB respectively. It is counted as two messages because 150 KB +120 KB = 270 KB, which is 1.08 times 250 KB.
- An integration flow has two receiver adapters. During runtime execution, messages are processed via both the receiver adapters and size of the messages is 300 KB and 220 KB respectively. It is counted as three messages because 300 KB + 220 KB = 520 KB, which is 2.08 times 250 KB.
- An integration flow is receiving a response for an outbound call. During runtime execution, messages are processed via one receiver adapter and the size of the request message is 150 KB and response message is 120 KB. It is counted as two messages because 150 KB (request) + 120 KB (response) = 270 KB, which is 1.48 times greater than 250 KB.
- An integration flow has two receiver adapters - one receiver adapter used in a request-reply step is receiving a response, and the other receiver adapter used to send messages to the target system without getting a response. During runtime execution, messages are processed via the 1st receiver adapter and the size of the request message is 50 KB and the size of the response message is 70 KB. Then, further transformation is applied and a message of 100 KB is sent via the 2nd receiver adapter to the target system. It is counted as one message because 50 KB (request from 1st receiver) + 70 KB (response to 1st receiver) + 100 KB (message to 2nd receiver) = 220 KB.

Special Notes or exceptions to above:

- The request size of the message received from the previous step by the receiver adapter is considered for size measurement. For example: If the size of the input message to receiver adapter is 120 KB and final outgoing message from receiver adapter over the wire is 130 KB, then 120 KB is considered for size measurements
- If a response is received via receiver adapter, the message size of both request and response messages processed by receiver adapter together is considered for the message size measurements
- In case of integration flows using splitter, each split outbound message via receiver adapter(s) is counted as separate message (subject to size and logic as mentioned above). Size of input message to splitter is 210 KB. The message is split to 3 messages. After further processing, the final size of each split message is 70 KB, and this is sent as input to receiver adapter. Total messages would be three messages of 70 KB each as per the message metering guidelines mentioned above.
- If you have configured your integration flow to retry message delivery during unavailability of receiver system or any other issues with the receiver system, each retried outbound message via receiver adapters is counted as a separate message (subject to size and logic as mentioned above).
- If multiple splitters are used in the same integration scenario, the splitter with the highest number of split messages via receiver adapter is considered for message count. For example, consider that your integration scenario has two splitters, splitter-1 and splitter-2. Spiltter-1 splits the message into 2 outbound messages while splitter-2 splits the message into 4 outbound messages. Here, we consider the splitter-2 for message count since it is splitting the message into 4 outbound messages.
- Outbound message size via JMS and Process Direct receiver adapters are not considered towards message count as these messages are transmitted within the Integration Suite tenant internally. JMS receiver adapter writes to JMS queues and Process Direct adapter forwards the messages to another iflow. As a rule of thumb, you can consider that messages processed via JMS and Process Direct receiver adapters is excluded from message metering.
- If a subprocess is performing outbound calls using a receiver adapter, it is counted with aforementioned criteria.

Note: The total number of message processing logs (MPL) that is shown in the Monitor view is not equal to the total number of messages. 1 MPL can contain "0" to "n" number of messages.

Limitation

Message size measurement is not available as of now. Until it gets available, every integration flow execution with 1 or more receiver adapters will be counted as 1 message irrespective of message size.

API Management

In the context of API Management, a capability of SAP Integration Suite, the API calls (& payload size) are considered for message metering.

For every API call via API Management, message size is measured. Each message size of up to 250 KB is calculated as 1 message. If message size is greater than 250 KB, the excess is counted in blocks of 250 KB to determine the total number of messages. In case of calls where a response is provided by the target system, message size of both request and response are considered. As a rule of thumb, you can determine the number of messages by dividing the aggregated message size by 250 and rounding up the answer to the next highest whole number.

Note: For API calls where there is no response, the incoming message to API Management are considered for size. For API calls that receive a response from the target system, the total size of incoming message to API Management and response from the target system are considered for message size.

Limitation: Message size measurement is not available as of now. Until it is available, every API call via API Management capability is counted as 1 message irrespective of size.

Edge Integration Cell

The messages processed by Edge Integration Cell are charged at 50%, meaning that 2 messages processed by Edge Integration Cell will be charged as 1 transaction. The metering of messages will follow the same framework that has been explained in Cloud Integration and API Management capabilities' section, depending on which functionality you are using in Edge.

Open Connectors

In the context of the Open Connectors capability of SAP Integration Suite, API calls (& payload size) are considered for message metering.

For every connector-based API call (including events and bulk) via Open Connectors capability of Integration Suite, message size is measured. Each message size of up to 250 KB is calculated as 1 message. If message size is greater than 250 KB, the excess is counted in blocks of 250 KB to determine the total number of messages. As a rule of thumb, you can determine the number of messages by dividing the aggregated message size by 250 and rounding up the answer to the next highest whole number.

Note: For API calls where there is no response, request message processed via the connectors or event feature of Open connectors capability is considered for size. For For API calls that receive a response from the target system, the total size request message and response message processed via the connectors or event connectors of Open Connectors capability are considered for message size.

Limitation: Message size measurement is not available as of now. Until it is available, every API call via connector and event features of the Open Connectors capability is counted as one message irrespective of size.

Event Mesh

In the context of Event Mesh capability, only outbound events are considered for message metering.

Any outbound event greater than 250 KB in size will be counted in multiples of 250 KB. For example, an event with size of 400 KB will be metered as 2 event messages.

100 event messages equal 1 transaction.

SAP-to-SAP Free Messages

SAP ships SAP-to-SAP integration scenarios for SAP Integration Suite (Cloud Integration capability) on SAP Business Accelerator Hub. This content provides an option to “Configure” the required fields (technical - host names, ports, security, etc. | business content specific – company code, subsidiary name, etc.) by selecting the “Configure” option. You can enter these values and deploy the integration flows.

Please note that the below criteria apply only to prepackaged integration scenarios shipped by SAP for SAP-to-SAP integration.

These SAP-to-SAP integration scenario flows are offered by SAP in two modes:

1. **Configure-Only Mode**

   In configure-only scenarios, you do not have an option to "Edit" the integration flow. You can only configure the exposed values by selecting the "Configure" button. All messages processed by this type of integration flows **are not deducted from your overall allocated monthly message quota.**

2. **Editable Mode**

   In these integration flows, in addition to the aforementioned "Configure" option, you also have the option to "Edit" the integration flows, if required. This is provided to help you extend the standard content by adding new steps in between or change it to fit your use-case.

While using this type of integration flows, here are some important points to consider:

1. If you do not "Edit" the integration flow and just use the "Configure" button to configure and deploy it, the messages processed by these flows **are not deducted** from your overall allocated message monthly quota, similar to how it is explained in point 1.
2. If you edit the integration flow by selecting the "Edit" button, the integration flow will be considered as customized. The messages processed by these flows **are counted** towards your overall allocated monthly message quota.

Extension Scenarios: *For SAP-to-SAP integration flow extension scenarios, if you use the pre/post exits that has been provided using the "Configure" option, the messages processed by that integration flow will not be reduced from your available message quota. The messages processed by the "extension" integration flow that you developed or the standard pre/post extension integration flows you modified will be reduced from your available message quota.*

Message Usage Report:

Analyze the number of messages processed by the active integration flows within a specific time interval and refer to SAP Help Documentation for more details.

IMPORTANT

- For integration scenarios shipped by SAP (Vendor: SAP):
  - Messages processed by SAP-to-SAP integrations are free of cost only if you use the integration packages without making any changes to them (by clicking on "Edit").
  - Message execution via SAP-to-Non-SAP and Non-SAP-to-Non-SAP integration packages are metered and counted towards your overall allocated monthly message quota.
- New integration scenarios that you create from scratch on your Cloud Integration tenant for SAP-SAP or SAP-Non-SAP integration is reduced from your monthly message quota allocation.
- Any content shipped by non-SAP vendors (including SAP-SAP integration) will be metered and reduced from your monthly message quota allocation.

This document is referenced by

SAP Note/KBA	Component	Title
2903776		SAP Integration Suite – Service Plans and Upgrade Paths